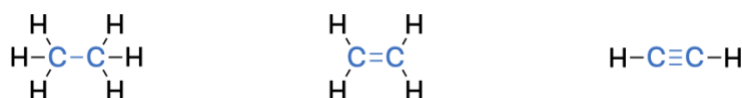


10.A Covalent Bonding I

Covalent Bonding. Nonmetal atoms bond with one another by sharing electrons in covalent bonds. In this chapter we'll learn how to describe and represent covalent bonding in molecules composed of nonmetals. We'll begin with some general considerations of covalent bonding, including the dependence of bond energy on internuclear distance.

- A **covalent bond** involves the sharing of valence electrons between two atoms (typically nonmetals). The two bonded atoms share ("co-") valence shells ("-valent").
 - Covalently bonded atoms may share two electrons (**single bond**), four electrons (**double bond**), or six electrons (**triple bond**).



- The energetics of covalent bonding involve a balance between electron-nucleus attraction and electron-electron repulsion.
 - The internuclear distance at the minimum of energy is called the **bond length** r_e .
 - The depth of the energy well is called the **bond dissociation energy** (D_e or BDE).

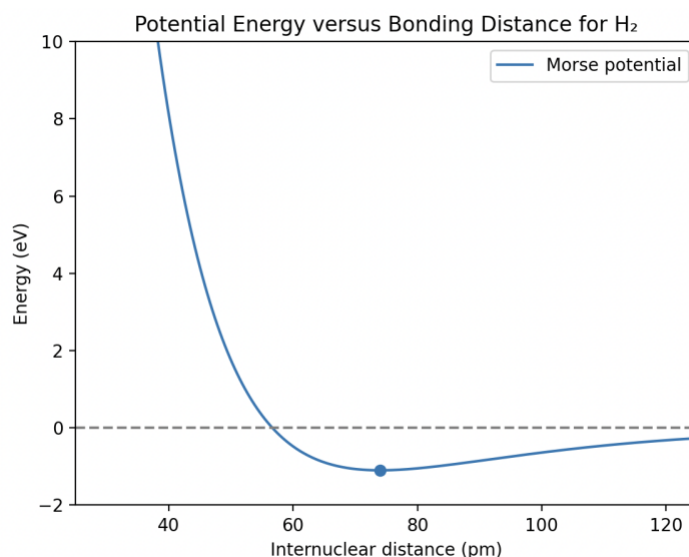
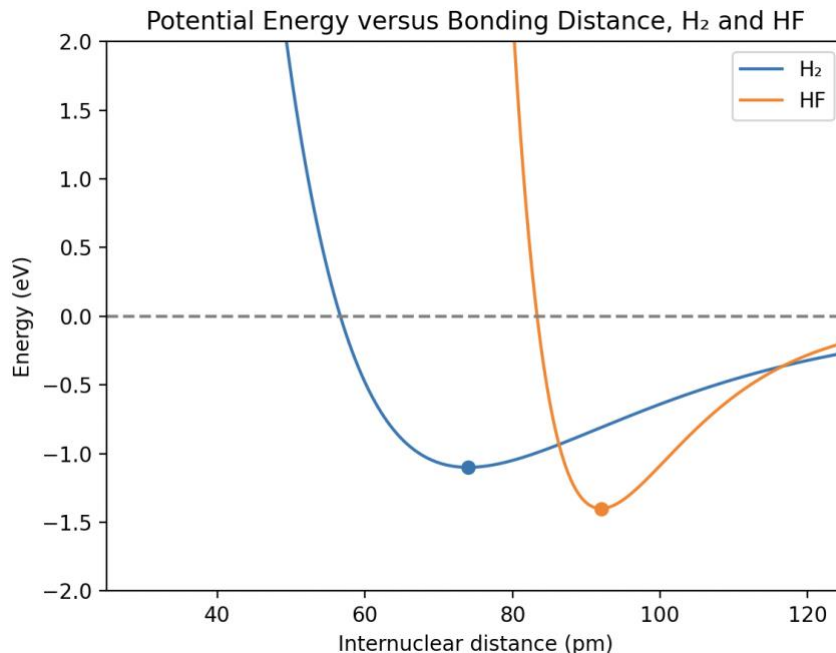


Figure. Energy as a function of the distance between the nuclei in an H₂ molecule.

10.1. The graph below shows potential energy as a function of internuclear distance for H_2 and HF . Determine the stronger of the two bonds and the shorter of the two bonds from the graph.



- A. The stronger bond is H-H and the shorter bond is H-H.
- B. The stronger bond is H-F and the shorter bond is H-H.
- C. The stronger bond is H-H and the shorter bond is H-F.
- D. The stronger bond is H-F and the shorter bond is H-F.

Electronegativity. We have one last periodic trend to discuss, which turns out to be convenient to bring up here. We have seen previously that the effective nuclear charge felt in the valence shell (Z_{eff}) is the underlying factor that accounts for ionization energy, electron affinity, and other periodic properties of the elements. The extent to which an atom attracts electrons to itself within molecules, called “electronegativity,” is also profoundly related to Z_{eff} . Electronegativity is an extremely important concept—you will apply it (and trends in it) repeatedly throughout your study of chemistry.

- **Electronegativity** is the capacity or ability of an atom within a molecule to attract electrons to itself.

- Deep conceptual correlations with Z_{eff} , ionization energy, and electron affinity
- Different quantitative scales have been developed for electronegativity.
 - Mulliken: the average of ionization energy and electron affinity.

$$\chi = \frac{IE + EA}{2}$$

- Pauling: hydrogen is arbitrarily set to 2.20 and differences in electronegativity between elements A and B are defined by the square root of the difference between the bond dissociation energy of AB and the average bond dissociation energy of AA and BB (all in eV).

$$|\chi_A - \chi_B| = \sqrt{D_e(AB) - \frac{D_e(AA) + D_e(BB)}{2}}$$

- Electronegativity increases left to right across a period and decreases down a group.
This is the most important periodic trend!
- More electronegative elements have greater electron density in covalent bonds (electrons are *unequally* shared!).
- Coincidentally, they also generally form fewer bonds and wind up on the periphery of Lewis structures.

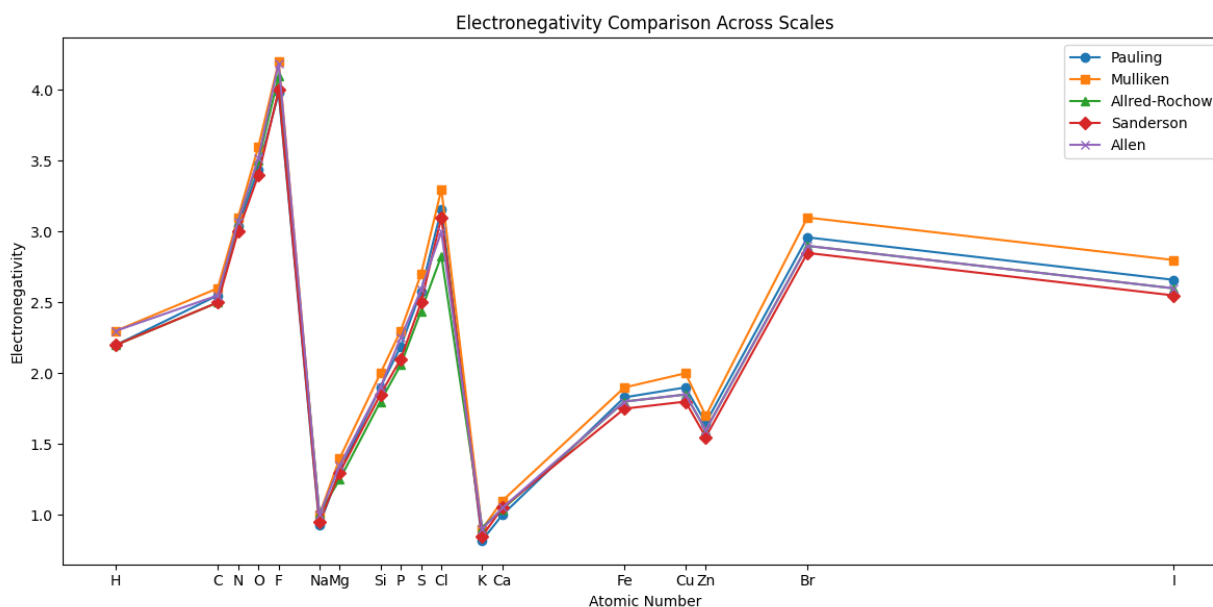


Figure. Electronegativities of the elements. Note the trends across a period and down a group.

10.2. Rank the elements carbon (C), nitrogen (N), oxygen (O), and fluorine (F) from least to most electronegative. Enter your ranking as a string of element symbols with no spaces or other delimiters (e.g., “CNOF”).

Bond Enthalpies and ΔH°_{rxn} . Bond enthalpy is a measure of covalent bond strength that is closely related to bond dissociation energy. We can use bond enthalpies together with knowledge of bonds made and broken in a reaction to estimate the enthalpy of reaction ΔH°_{rxn} . This approach is an alternative to the one based on enthalpies of formation that we’ve already seen; it can be more convenient when we don’t know an enthalpy of formation but we have access to relevant bond enthalpies.

- **Bond enthalpy (bond dissociation enthalpy, ΔH°_{BD}),** a measure of the strength of covalent bonds, is defined as the enthalpy change when 1 mol of bonds is broken **homolytically**.



- Usually reported in kilojoules per mole (kJ/mol) as an average for cleavage of X–Y bonds in many different molecules
- BDEs are always positive because **bond cleavage is endothermic** and **bond formation is exothermic**.
- Stronger bonds have higher BDEs. Triple bonds are stronger than double bonds, which are stronger than single bonds. Stronger bonds are generally shorter than weaker bonds (but not always—see above!).
- Bond enthalpies can be used with a list of bonds made and broken in a reaction to estimate ΔH°_{rxn} .

reactants $\xrightarrow{\text{cleave all bonds broken}}$ fragments $\xrightarrow{\text{form all bonds made}}$ products

$$\Delta H^\circ \approx \sum_i^{\text{bonds broken}} \Delta H^\circ_{BD,i} - \sum_j^{\text{bonds made}} \Delta H^\circ_{BD,j}$$

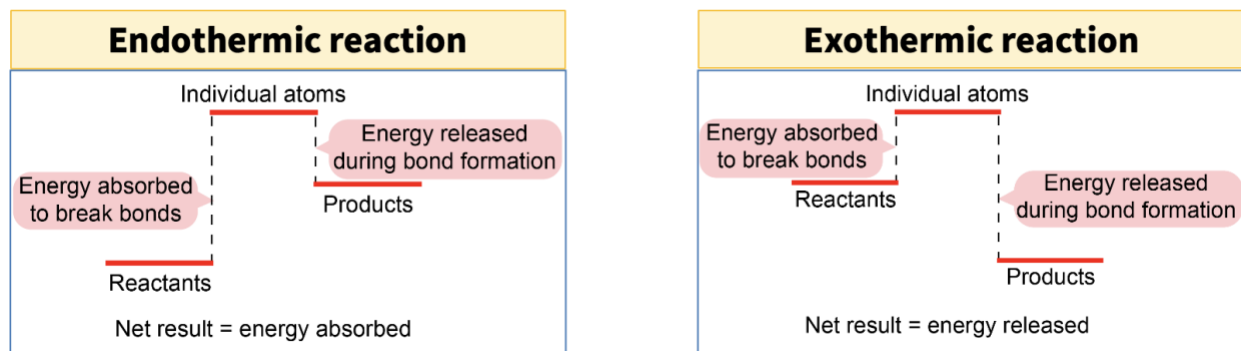
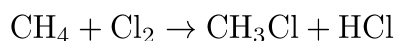


Figure 10.19. Bond enthalpies in endothermic and exothermic reactions.

10.3. Use bond enthalpies to estimate the enthalpy of reaction for the gas-phase chlorination of methane to give methyl chloride and hydrogen chloride. Relevant bond enthalpies are listed below. Enter your response in kilojoules per mole (kJ/mol) to the nearest 1 kJ/mol with appropriate sign.



C-H: 413 kJ/mol

Cl-Cl: 239 kJ/mols

C-Cl: 339 kJ/mol

H-Cl: 427 kJ/mol

10.B Covalent Bonding II

Introduction to Lewis Structures. Lewis structures are used to represent molecules containing covalently bonded atoms. Their virtue is their simplicity—we can quickly generate a Lewis structure given the connectivity of a molecule. Although they are a simplification of the true quantum-mechanical electron density in molecules, they are very frequently a reliable representation. We'll start by introducing some general concepts for Lewis structures.

- We represent the structures of molecules using element symbols, lines, and dots in a **Lewis structure**. *Only* valence electrons are included in Lewis structures but all valence electrons should be accounted for.
- Bonds may be single, double, or triple (see above).
- **Nonbonding lone pairs** are valence electrons that do not participate in bonding. They are represented as dots near the element symbol to which they “belong.”

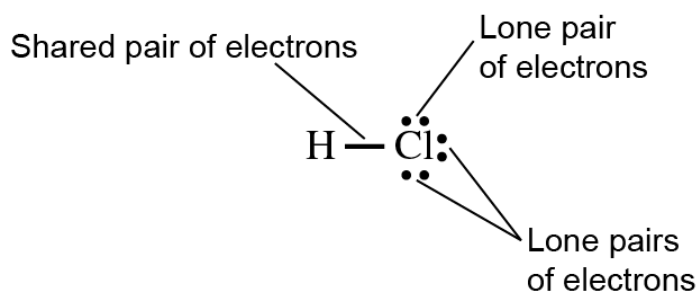


Figure 10.2. A Lewis structure for hydrogen chloride.

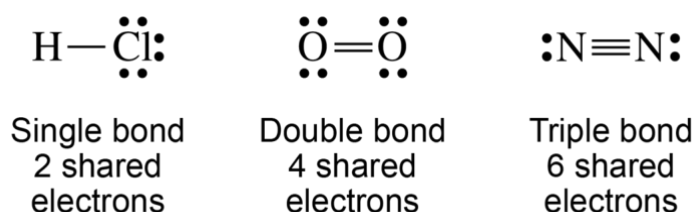


Figure 10.3. Examples of single, double, and triple bonds in Lewis structures.

- It has been observed that atoms frequently form enough bonds to achieve a total electron count of eight. This guideline is known as the **octet rule**.
 - The octet rule is **particularly strong** for N, O, and F. An important exception is hydrogen (H), which always forms just one bond.

- *Why eight?* For a p -block element, this count corresponds to a full atomic valence shell (ns^2np^6) and the maximum number of what we'll later call bonding molecular orbitals.
- Note that the total number of electrons in a Lewis structure must equal the total number of valence electrons in the atoms of the molecule.
- In the Lewis structure of carbon dioxide, note the total number of electrons and the octet at each atom. To evaluate octets, we **double count** shared electrons.



Figure 10.4. Lewis structure of carbon dioxide highlighting the octets at oxygen (a) and carbon (b).

- Drawing a Lewis structure given a molecular formula...
 1. Determine the total number of valence electrons. For cations, subtract one electron for each positive charge. For anions, add one electron for each negative charge.
 2. Draw a **skeletal structure** of the molecule or ion, arranging the atoms around a central atom. (Generally, the element with smallest Z_{eff} should be placed in the center.) Connect each atom to the central atom with a single bond.
 3. Distribute the remaining electrons as lone pairs on the peripheral atoms (except hydrogen), completing an octet around each atom.
 4. Place all remaining electrons on the central atom.
 5. Rearrange the electrons of the outer atoms to make multiple bonds with the central atom to obtain octets wherever possible.
 6. Assign **formal charges**, making sure no atom has charge greater than ± 1 and the sum of all charges equals the total charge of the molecule or ion.
- *Always* carry out a sanity check on your structure.

Example. Draw a Lewis structure for H_2O_2 .

Example. Draw a Lewis structure for CH_3OCH_3 . Think carefully about how the atoms are connected. How can the skeletal structure be inferred from the way the formula is written?

10.4. Draw a Lewis structure for SiCl_4 . How many nonbonding lone pairs does the structure contain? Enter your response as a numeral.

- **Formal charge** is based on the count of valence electrons “belonging” to each atom in the structure. It provides a *rough* answer to the question “where is the charge?”
 - May resemble the more realistic atomic partial charges...
 - ...*but* often does not due to resonance delocalization (more soon).

$$\text{formal charge} = \text{valence electrons in neutral atom} - [\text{nonbonding electrons} + \frac{1}{2}(\text{bonding electrons})]$$

- Over time, you should develop the ability to infer formal charge from common **bonding patterns** of the *p*-block elements—*NOT* by using the formula above. The formula is just an “analytical failsafe” for unfamiliar cases.

Example. Draw a Lewis structure for carbonate anion (CO_3^{2-}).

10.5. The carboxylate group is CO_2^- . Draw a Lewis structure for HCO_2^- , which includes this group. This structure will include at least one atom with formal charge. Which answer choice best describes the atom(s) with formal charge?

- A. Hydrogen has a formal charge of -1 .
 - B. Carbon has a formal charge of -1 .
 - C. One oxygen atom has a formal charge of -1 .
 - D. Both oxygen atoms have formal charges of -1 .
 - E. All atoms in this structure are neutral.
-

Exceptions to the Octet Rule. The octet rule is more of an octet “guideline.” We’re already familiar with the idea that hydrogen only bears two electrons in Lewis structures; this is one important exception. Here, we’ll see that in the third period and below, it is common for nonmetal atoms to bear more than eight electrons (a so-called “expanded octet”). Group 13 elements such as boron, aluminum, and gallium are frequently found in molecules with only six total electrons, since they are formally neutral in this situation.

- The octet rule is best applied to second-row elements in groups 14 – 17 (C, N, O, F). There are a few general exceptions to the “rule.”
 - **The duet rule.** Hydrogen forms only one bond to achieve a “duet” ($1s^2$).
 - **Incomplete octets.** Group 13 elements such as B, Al, and Ga only form three bonds to remain formally neutral. For a similar reason, Be in group 12 forms only two bonds (but Be rarely bonds covalently).
 - **Expanded octets.** Elements in periods 3 and below can and often do bear more than eight electrons. This is common when an element is linked to more atoms than its period 2 congener suggests it “should” be.

Example. Consider the Lewis structures of H_2SO_4 , H_2S , and H_2O . Where is the octet rule violated?

Example. The nitro group -NO_2 is an important electron-withdrawing group in organic molecules. Draw a valid Lewis structure for CH_3NO_2 , which contains the nitro group. Your structure will include nonzero formal charges; make sure you understand why!

10.6. Dimethyl sulfate, $(\text{CH}_3\text{O})_2\text{SO}_2$, was used as an alkylating reagent in the days of yore before it was found to be extremely toxic. Draw a valid Lewis structure for this molecule and indicate where it contains an expanded octet. What is the total electron count at the atom that has an expanded octet?

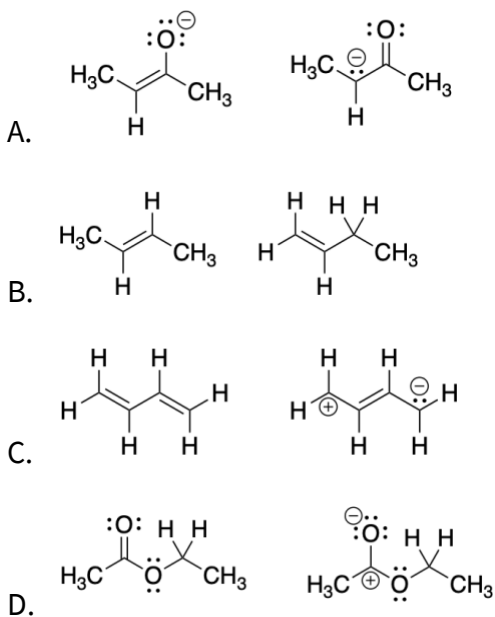
Hint: to figure out the connectivity, compare to sulfuric acid above.

10.C Covalent Bonding III

Delocalized Electrons and Resonance. Lewis structures are fundamentally *localized* representations. Electrons are either “sitting on” a single atom or shared between two atoms. But the quantum nature of electrons tells us that this is a simplification—in reality, electrons can “stretch their legs” over the entirety of the molecule. While true, in most molecules, electrons can be very well approximated as localized and Lewis structures retain their usefulness. In molecules containing profoundly delocalized electrons, we use multiple Lewis structures to represent the delocalization. We’ll see how that works here.

- In Lewis structures, electrons must be located either on an atom or between two atoms. But in many molecules, electrons are **delocalized** over three or more atoms, meaning they cannot be represented with only one Lewis structure!
- Molecules that exhibit electron delocalization are represented using multiple Lewis structures called **resonance forms**.
- Resonance forms differ *only* in the placement of multiple bonds and lone pairs—**nuclei and single bonds do not move**.¹

10.7. Which of the following pairs of structures are *NOT* related as resonance forms? Carbon atoms are located at the intersections of bonds without an atom label.



¹ The underlying idea here is that only pi electrons experience delocalization. The meaning of this will become clearer in the next chapter.

- **Resonance hybrid:** the weighted average of all valid and significant resonance forms for a molecule. Valid structures have reasonable formal charges (-1, 0, or +1) and “respect the octet guideline.”
 - The resonance hybrid resembles the “true” structure of the molecule.
 - The weights, known as **contributions**, depend on “how important” the structure is to the resonance hybrid.

Example. Draw a valid Lewis structure for the nitrite anion (NO_2^-). Then, draw an alternative resonance form and the resonance hybrid for this molecule.

10.8. Let’s revisit the carbonate anion (CO_3^{2-}). Draw structures for the three valid resonance contributors for this molecule and the resonance hybrid. Which statement best describes the negative charge in this molecule?

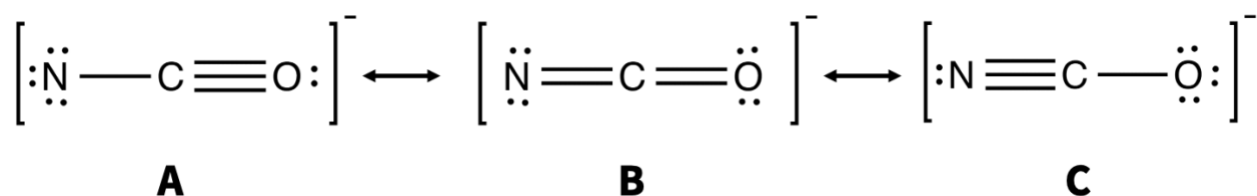
- A. The negative charge is localized on carbon.
- B. The negative charge is localized on a single oxygen atom.
- C. The negative charge is shared unequally by the oxygen atoms.
- D. The negative charge is shared equally by the oxygen atoms.

-
- *How do we rank the relative contributions of resonance forms?*
There are three ideas to consider...

1. Fewer and smaller formal charges are preferable to larger charges.

- Avoid like charges (++) or (--) on adjacent atoms and separate like charges as much as possible. Keep opposing charges (+-) as close as possible.
- Negative formal charge should be located on more electronegative atoms.
Positive formal charge should be located on less electronegative atoms.

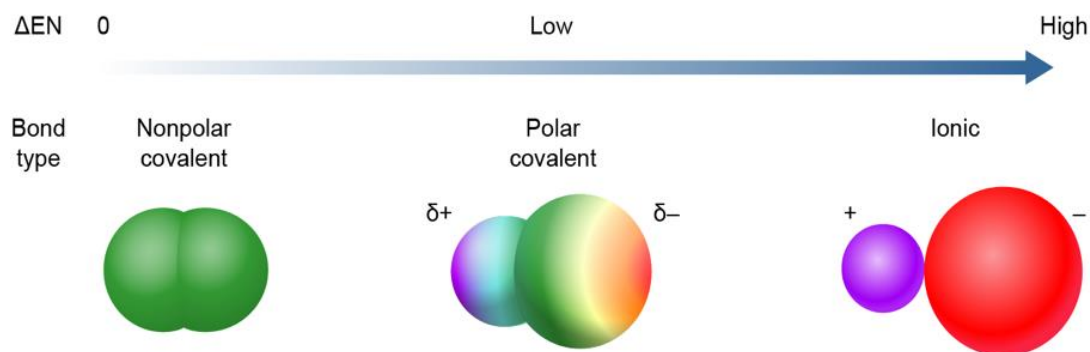
10.9. Assign formal charges to all the atoms in the three resonance forms of NCO^- labeled A – C below. Rank the structures from greatest (1) to smallest (3) contributions to the resonance hybrid. Enter your ranking as a string of letters with no delimiters between them (e.g., “ABC”).



Polar Bonds and the Bonding Continuum. Electronegativity, formal charges, and resonance all point to the idea that in many covalent bonds, electrons are not shared equally. Covalent bonds with profoundly unequal sharing of electrons are said to be “polar,” since they contain a positive end and a negative end. We can imagine chemical bonds on a kind of continuum of polarity from “purely covalent” bonds such as H–H to “purely ionic” bonds such as $\text{Cs}^+ \text{F}^-$. Electronegativity guides our thinking when locating bonds on this continuum.

-
- The figure displays electrostatic potential maps and Lewis structures for two molecules. A vertical color scale bar is positioned between the two molecular representations, ranging from red at the top (labeled 'Most negative (δ-)') to blue at the bottom (labeled 'Most positive (δ+)').
- (a) Hydrogen fluoride (HF): The electrostatic potential map on the left shows a large red region on the fluorine atom and a small blue region on the hydrogen atom. Below the map is the Lewis structure: $\delta^+ \text{H} - \ddot{\text{F}} : \delta^-$.
- (b) Fluorine (F₂): The electrostatic potential map on the right shows two large red regions, one on each fluorine atom. Below the map is the Lewis structure: $:\ddot{\text{F}} - \ddot{\text{F}}:$.

- The difference in electronegativity between two atoms in a bond ($\Delta\chi$) can be used to classify bonds on a continuum from fully covalent to fully ionic.



- Polar bonds can be envisioned as equal and opposite charges (+1 and -1) separated by a distance r .
- The associated **dipole moment (μ)** is a measurable quantity that we can represent as a **dipole vector** from the +1 end to the -1 end.

- The greater $\Delta\chi$, the greater the dipole moment and the longer the dipole vector.



- We will soon use bond dipole vectors to evaluate the polarity of entire molecules.

Chapter Summary. Some of the amazing things we can now do with covalent bonding...

1. Use graphs of potential energy as a function of internuclear distance to evaluate the lengths and strengths of covalent bonds.
2. Define and apply the concept of electronegativity; apply periodic trends in electronegativity to rank a series of elements.
3. Estimate enthalpy of reaction using bond enthalpies and a list of bonds made and broken.
4. Draw Lewis structures for simple small molecules and ions given a molecular formula and, where necessary, information about connectivity.
5. Draw and interpret resonance forms for molecules exhibiting electron delocalization.
6. Combine resonance forms to generate the resonance hybrid, an average representation of molecular structure.
7. Apply differences in electronegativity to classify a covalent bond as polar or nonpolar and draw bond dipole vectors.